Amenities Magnet

Identifying Key Drivers of Residential Rental Prices in Germany

Team 146 · CSE6242: DVA · Spring 2026
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Our Approach

What are we doing?

Predicting residential rents in Germany + visualizing which amenities drive prices in each city

Why does it matter?

Renters, investors & policymakers lack transparency into what drives rent differences across cities

What's the gap?

Existing models output a price as a black box; no interactivity, no city-specific driver insight

Our Solution

1. **Advanced Modelling**

   Captures nonlinear interactions beyond simple regression approaches

2. **City-Specific**

   Provide separate models per city to capture drivers differing by market and region

3. **Interactive Dashboard**

   Streamlit-based app explores feature importance across cities

4. **Explainability First**

   SHAP + feature rankings show why a property costs what it does

Beneficiaries & Expected Impact

Tenants

Interrogate if listings are fairly priced and understand which amenities justify price differentials

Investors

Target highest-ROI features for design, renovation, and purchasing decisions

Developers

Guide amenity investment decisions with data-backed feature importance rankings

Policymakers

Understand how infrastructure & neighborhood factors drive affordability across cities

Broader Impact

Reduces information asymmetry between renters, owners, & investors

Leads to more transparent pricing

Leads to fairer housing market transactions.

If widely adopted, improves rental income valuation accuracy and overall market efficiency.

Literature Review & USP

PRIOR WORK

[Ho et al. 2021]

Tree ensembles outperform linear models but rarely translated into user-facing tools

[Lenaers & De Moor 2023]

XAI techniques improve rent model transparency, but explanations typically static

[Lorenz et al. 2023]

City-level German ML: size & age are primary drivers — but no cross-city comparison

[Metz-Peeters 2025]

Spatial heterogeneity in German cities — price drivers shift across markets

[Tapia et al. 2025]

SHAP + ensemble AVM outperforms hedonic baselines; accuracy + explainability pair well

OUR INNOVATIONS

Algorithm

XGBoost with city-stratified training; SHAP values for local feature attribution

Cross-City

Separate models per city reveal how drivers shift across German markets

Multi-Model

Benchmarks XGBoost vs RF vs baseline — no single model assumed best

Visualization

Interactive Streamlit dashboard — users explore feature importance by city, not just static charts

Plan of Activities and Goals

Task 1: 01Mar – 15Mar

Exploratory Data Analysis

- Clean ImmoScout24 data
- Exploratory analysis
- Feature engineering

Task 2: 01Mar – 15Mar

Modelling Building

- XGBoost, RF, baseline
- Cross-validation
- Evaluate per-city accuracy

Task 3: 15Mar – 29Mar

Dashboard

- Extract SHAP values
- Build Streamlit app
- City comparisons UI

Task 4: 29Mar - 24Apr

Integration

- System integration
- Usability testing
- Final deliverables

GOALS

Midterm:

Data prepared, tuned models beat baselines and initial feature rankings

Final:

Accurate predictions and stable rankings presented through working interactive dashboard

Risks, Cost & Summary

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Generalization

Models may perform unevenly across cities: mitigated by city-stratified training & multi-model benchmarking

DQ

Data Quality

ImmoScout24 may have missing fields or erroneous data: addressed in Week 1 cleaning pipeline

UX

User Interface Simplicity

Balancing dashboard complexity vs. usability: iterative usability testing in Week 4

Associated Costs….

$0

Free: Kaggle data & Streamlit software

The Payoff

A tool that predicts rent and shows users logically why. Thus, empowering tenants, investors, and policymakers with transparent, city-specific insights.